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Air pollution removal by urban trees and shrubs in the United States

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1. Abstract & Scope

Air pollution is a major threat in US cities. This study modeled the annual removal of air pollutants (PM_{2.5}, O₃, NO_x, SO_x, CO) by urban trees across the United States. Combining field surveys from 55 cities with hourly weather and pollution datasets in the Urban Forest Effects (UFORE) model, we estimated that urban trees remove 711,000 metric tons of pollutants annually, with PM_{2.5} removal representing a major fraction (215,000 metric tons). While average local air quality improvement was small (0.51%), local PM reduction in dense forest patches exceeded 10%.

2. Methodology & Experimental Parameters

UFORE model application combining field plot data, leaf area index (LAI), hourly weather observations, and EPA pollution monitor telemetry.

3. Core Findings & Data Synthesis

Urban trees provide massive economic value through air filtration, with pollution removal rates directly proportional to canopy coverage and growing season length.

4. Strategic Relevance to JU and Kolkata Planning

Establishes macro-scale economic and health value of trees, reinforcing the need to convert city green zones into active air filtration networks.

